NuMagSANS Tutorial 4 - Helical Magnet

NuMagSANS Tutorials - Michael P. Adams

Version: July 7, 2026

Goal

Build and inspect one helical magnetic sphere on a local workstation.

- Use the SystemDesigner sphere workflow.
- Materialize one spherical local object.
- Write one magnetic vector-field file, `m_1.csv`.
- Inspect the generated field with a PyVista diagnostic plot.

Helical State

The local magnetic state is a transverse helix propagating along the local z-axis.

$$\mathbf{m}(z) = \mathbf{e}_x \cos(kz + \varphi_0) + \chi \mathbf{e}_y \sin(kz + \varphi_0)$$

- `k` controls the pitch.
- `chi` controls the chirality.
- Global orientation distributions can be added later through `RotData`.

Python File

Create a file named `helical_single_sphere.py`.

```
import math
from pathlib import Path

from NuMagSANS.SystemDesigner.AssemblyAnalyzer import plot_magnetization_file
from NuMagSANS.SystemDesigner.Workflows import write_spherical_replication_vectorfield_sweep
```

Parameters

Use one lattice point per nanometer for a clear first look.

```
output_dir = Path("HelicalSphere_R20")

radius_nm = 20.0
lattice_constant_nm = 1.0
pitch_nm = 20.0
chirality = 1.0
phase = 0.0
```

Generate Real-Space Data

The workflow writes one local object and one magnetization file.

```
summary = write_spherical_replication_vectorfield_sweep(
    output_dir=output_dir,
    R=radius_nm,
    a=lattice_constant_nm,
    atomtype="Fe",
    n_replications=1,
    field_parameter_cases={
        "field_type": "transversal_helix",
        "k": 2.0 * math.pi / pitch_nm,
        "chirality": chirality,
        "phase": phase,
    },
    n_rotdata=1,
    beta_min=0.0,
    beta_max=0.0,
    rotation_seed=1,
)
```

Diagnostic Plot

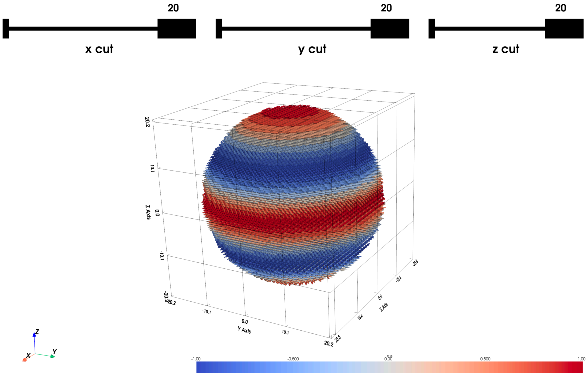
The plot is intended for local inspection, not for HPC production runs.

```
mag_file = (
    Path(summary["real_space_dir"])
    / "MagData"
    / "Object_1"
    / "m_1.csv"
)

plot_magnetization_file(
    mag_file,
    vector_scale=1.6,
    center_vectors=True,
    seed=1,
    show_points=False,
    color_by="mx",
    cmap="coolwarm",
    enable_cut_sliders=True,
    arrow_tip_length=0.63,
    arrow_tip_radius=0.15,
    arrow_shaft_radius=0.06,
)
```

Result

The generated vector field shows one transverse helical period across the sphere diameter.



Run Locally

Run the script from the NuMagSANS repository root.

```
python docs/tutorials/tutorial_4_workflows/python/helical_single_sphere.py
```

The generated data are written to `HelicalSphere_R20/RealSpaceData`.

Generated Data

The workflow writes the real-space input data for one local object.

- `HelicalSphere_R20/RealSpaceData/`
- `Local_Objects/Object_1/pos.csv`: lattice positions inside the sphere.
- `Local_Objects/Object_1/atomtype.csv`: atom labels for these positions.
- `Local_Objects/Object_1/meta.csv`: object metadata.
- `MagData/Object_1/m_1.csv`: helical magnetization vector field.
- `RotData/RotData_1.csv`: orientation data for the object replication.

Run NuMagSANS

Use the generated data as input for a NuMagSANS simulation.

```
python docs/tutorials/tutorial_4_workflows/python/run_numagsans_helical_sphere.py
```

The script writes `HelicalSphere_R20/NuMagSANSInput_helical_sphere.conf`, starts NuMagSANS, and stores CSV output in `HelicalSphere_R20/NuMagSANS_Output`.